Voice Biometric Intelligence & Voice Cloning Detection

COMPREHENSIVE TECHNICAL & VIVA DEFENSE GUIDE (BiLSTM, FFT, MFCC) — B.Tech CSE (23CSE750)

Project: Voice Biometric Intelligence & Voice Cloning Detection

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Topic 1: BiLSTM (Bidirectional Long Short-Term Memory)

1.1 What is BiLSTM and how does it work?

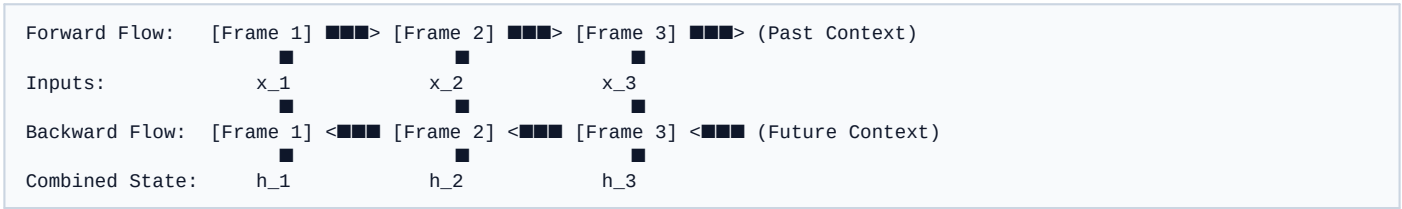
A **Bidirectional Long Short-Term Memory (BiLSTM)** is a deep learning recurrent network that analyzes sequential data in **both directions simultaneously**:

1. **Forward Pass:** Reads the audio features from the start to the end ($t = 1 \rightarrow T$).
2. **Backward Pass:** Reads the audio features from the end to the start ($t = T \rightarrow 1$).

At each time step t, the network combines the forward hidden state $\overrightarrow{h}_t$ and backward hidden state $\overleftarrow{h}_t$:

$$h_t = [\overrightarrow{h}_t \,\|\, \overleftarrow{h}_t]$$

This allows the model to understand what sound was spoken **before** and what sound comes **after** any given moment.



1.2 What is LSTM, and why is it called Long Short-Term Memory?

Standard Recurrent Neural Networks (RNNs) suffer from the **vanishing gradient problem**: when processing long audio sequences, they forget earlier sounds.

LSTM fixes this by introducing an internal memory cell governed by 3 gates:

- **Forget Gate (f_t):** Decides what irrelevant information to discard.
- **Input Gate (i_t):** Decides what new acoustic information to store.
- **Output Gate (o_t):** Decides what information to output to the next layer.
- **Cell State (C_t):** A continuous memory track that carries information across hundreds of time steps without degrading.

Why is it called "Long Short-Term Memory"?

- **Short-Term Memory:** In neural networks, the hidden state h_t is temporary and changes rapidly at each time step.
- **Long-Term Memory:** The model weights learned during training are permanent.

• **LSTM:** By adding the **Cell State**, the network keeps its **short-term memory active over long durations**. Hence, *Long Short-Term Memory*.

1.3 What does "Bidirectional" mean in BiLSTM?

In speech, articulation is continuous. How we pronounce a vowel is physically influenced by:

1. The sound we just made (**past sound**).
2. The sound our mouth is already preparing to make next (**anticipatory co-articulation / future sound**).

A unidirectional model only hears the past. A **bidirectional** model processes the entire phrase in both directions, giving it complete temporal context.

1.4 How does BiLSTM learn patterns from audio signals?

An audio signal is sliced into sequential frames (e.g., 301 frames of 120-dimensional MFCC vectors for a 3-second recording).

• The BiLSTM tracks how vocal resonance (formants) change over time.

• It learns the natural speed, acceleration, and rhythm of human speech.

• It maps these temporal patterns into a single compact **voiceprint embedding vector**.

1.5 Why are we using BiLSTM in our project?

1. **Models Natural Speech Dynamics:** Accurately learns smooth phoneme transitions.
2. **Detects Unnatural AI Transitions:** AI-generated speech often has subtle frame-to-frame stitching errors or flat pitch curves that BiLSTM identifies.
3. **Works Well on Limited Data:** Unlike massive Transformers, BiLSTM trains stably on small-to-medium datasets without needing tens of thousands of samples.
4. **Lightweight & Fast:** Runs inference in under 50 ms on standard consumer laptop CPUs.

1.6 Why choose BiLSTM instead of other models?

• **vs. Standard LSTM / GRU:** Unidirectional LSTM only looks backwards and misses future context. BiLSTM achieves higher recognition accuracy.

• **vs. Vanilla RNN:** Standard RNNs cannot handle sequences longer than 15–20 steps due to vanishing gradients.

• **vs. 2D CNN:** CNNs are great for local image patches, but struggle with long-term sequential dependencies spanning multiple seconds.

• **vs. Transformers (Conformer / AST):** Transformers require huge training datasets and heavy GPUs; on small student datasets, Transformers easily overfit.

1.7 What specific advantages does BiLSTM offer for detecting AI-generated voices?

1. **Unnatural Pitch Jitter:** Real human voices have natural micro-variations (jitter and shimmer). Cloned voices often sound robotic or unnaturally smooth.
2. **Frame Boundary Discontinuities:** Neural vocoders (like HiFi-GAN) synthesize speech in chunks, leaving tiny temporal boundary glitches that BiLSTM detects.
3. **Prosody Tracking:** AI voices often misplace stress or pitch emphasis across whole sentences. BiLSTM evaluates rhythm across the entire 3-second window.

1.8 What are the limitations and computational requirements of BiLSTM?

• **Sequential Nature:** Cannot be fully parallelized during training because frame t depends on frame $t-1$.

• **Memory Consumption:** Storing forward and backward states for very long audio (>30 seconds) uses significant RAM.

• **Hardware Requirement:** Moderate. Can be trained easily on any consumer laptop (4 GB RAM, dual-core CPU or basic GPU) in under 15 minutes.

1.9 Is BiLSTM suitable for real-time voice detection? Does it introduce latency?

Yes, for chunked real-time processing; No, for instantaneous sample-by-sample streaming.

• **Why latency exists:** Because the backward LSTM needs to read from the end of the phrase back to the start, the system must wait until the full audio chunk (e.g., 2 to 3 seconds) is recorded.

• **In practice:** This 2–3 second buffer delay is completely acceptable for biometric authentication turnstiles, call verification, and upload inspection. Once the buffer is full, inference takes less than 50 ms.

1.10 When might BiLSTM perform poorly or fail?

1. **Very Short Audio (< 0.5 seconds):** Not enough time frames for recurrent memory to build up context.
2. **Heavy Echo / Reverberation:** Room reflections mix past and future sound waves, confusing the bidirectional tracking.
3. **State-of-the-Art Diffusion Voice Clones:** Cutting-edge models (e.g., ElevenLabs latest models) generate very smooth transitions that can mimic human temporal flow closely.

Topic 2: FFT (Fast Fourier Transform)

2.1 What is FFT, and how does it work?

The **Fast Fourier Transform (FFT)** is an efficient computer algorithm used to calculate the **Discrete Fourier Transform (DFT)** of a digital signal.

Instead of calculating every frequency step one by one ($O(N^2)$ complexity), FFT uses a **divide-and-conquer strategy** (Cooley-Tukey algorithm):

• It splits a block of N audio samples into even and odd halves.

• It recursively computes their frequencies and recombines them using mathematical symmetry.

• This reduces the computational cost to $O(N \log_2 N)$—about **50 to 100 times faster** than direct calculation.

2.2 What is the Fourier Transform, and why is it important in audio?

Sound is recorded as a **time-domain signal**: amplitude (air pressure) changing over time. But looking at raw amplitude cannot tell you what notes or voice frequencies are present.

The **Fourier Transform** mathematically breaks down a complex audio wave into its individual pure sine wave frequencies.

It answers the question: *"Which individual pitch frequencies and loudness levels make up this sound?"*

2.3 Difference between DFT and FFT

• **DFT (Discrete Fourier Transform):** The mathematical equation that maps discrete digital time samples into discrete frequency values.

• **FFT (Fast Fourier Transform):** The fast software algorithm used to compute the DFT quickly on computers.

(DFT is the formula; FFT is the fast algorithm).

2.4 What information does FFT extract from an audio signal?

1. **Frequencies:** What tones are present (in Hertz, Hz).
2. **Magnitude (Amplitude):** How loud each frequency is.
3. **Phase:** The starting angle/alignment of each wave component.
4. **Harmonics:** The fundamental voice pitch (F_0) and its integer multiples ($2F_0, 3F_0, \dots$).

2.5 How does frequency and spectral analysis help detect AI-generated voices?

1. **High-Frequency Cutoffs:** Many cloning tools synthesize audio at low sample rates (16 kHz or 22 kHz) and sharply filter out everything above 8 kHz. FFT reveals this unnatural "brick-wall" cutoff immediately.
2. **Vocoder Noise Spikes:** Synthesis tools can leave faint, regular frequency spikes at fixed intervals due to neural network upsampling layers.
3. **Unnatural Harmonic Stability:** Real human vocal cords vibrate with slight irregularities. Cloned voices often produce perfectly rigid harmonic lines.

2.6 Why are we using FFT in our project?

In our project, FFT is **not a standalone classifier**. It is the essential mathematical engine that converts raw time-domain audio samples into frequency spectra so that Mel filterbanks and MFCCs can be calculated.

2.7 Difference between FFT and STFT (Short-Time Fourier Transform)

- **Standard FFT:** Run on the **entire** audio file at once.
- *Limitation:* It tells you what frequencies existed, but loses **when** they happened! (Speech changes every 10 milliseconds).
- **STFT (Short-Time Fourier Transform):** Slices the audio into small overlapping windows (e.g., 25 ms windows every 10 ms) and runs an FFT on each window.
- *Result:* Creates a **Spectrogram** that shows **Time vs. Frequency vs. Energy**. This is ideal for speech analysis.

2.8 Can FFT alone reliably distinguish real human voices from cloned voices?

No.

1. A single FFT has no temporal order—it mashes all words into a static frequency histogram.
2. Modern AI voice clones reproduce standard frequencies very convincingly.
3. To catch an AI voice, you need both frequency information (from FFT/MFCC) and temporal sequence information (from BiLSTM).

Topic 3: MFCC (Mel-Frequency Cepstral Coefficients)

3.1 What are MFCCs, and what does the name mean?

Mel-Frequency Cepstral Coefficients (MFCCs) are a compact set of numbers that represent the overall shape of the human vocal tract during speech.

Breakdown of Terms:

- **Mel:** From the *Mel Scale* (melody), an auditory scale that mimics how the human ear hears pitch.
- **Frequency:** Based on spectral frequency analysis.
- **Cepstral:** An anagram of *Spectral*. The cepstrum is calculated by taking the inverse transform of the *logarithm* of the spectrum.
- **Coefficients:** The resulting numerical values (e.g., 40 numbers per frame) describing the vocal envelope.

3.2 Step-by-Step Process of Extracting MFCCs

```
Raw Audio Waveform (16 kHz)
       ■
       ▼
[Step 1: Pre-Emphasis] ■■■■■■■■> High-pass filter boosts weak high frequencies: y[n] = x[n] - 0.97·x[n-1]
       ■
       ▼
[Step 2: Framing & Windowing] ■■> Sliced into 25 ms frames with 10 ms hop; multiplied by Hann Window
       ■
       ▼
[Step 3: Fast Fourier Transform]> 512-point FFT converts each frame into a Power Spectrum |X[k]|²
       ■
       ▼
[Step 4: Mel Filter Bank] ■■■■■■■> 64 overlapping triangular filters group frequencies like the human ear
       ■
       ▼
[Step 5: Logarithm] ■■■■■■■■■■■■■> Log of energy values separates vocal tract shape from vocal cord pitch
       ■
       ▼
[Step 6: Discrete Cosine (DCT)] > Compresses 64 values into 40 clean, decorrelated MFCC coefficients
       ■
       ▼
[Step 7: Dynamic Deltas] ■■■■■■■■> Computes 40 Velocity (Δ) and 40 Acceleration (ΔΔ) coefficients
       ■
       ▼
Final Output: 120 numbers per 10 ms frame (Shape: 301 frames × 120 features for 3 seconds)
```

3.3 What is the Mel Scale, and why is it used?

Human ears do not perceive pitch in a straight linear line:

- We can easily hear the difference between 200 Hz and 300 Hz (low pitch).
- We can barely tell the difference between 7000 Hz and 7100 Hz (high pitch).

The **Mel Scale** is a mathematical formula that models this biological perception:

$$m = 2595 · \log_{10}≤ft(1 + \frac{f}{700}\right)$$

It spaces filter banks closely together at low frequencies and wider apart at high frequencies, matching how human hearing actually works.

3.4 How do MFCCs represent human speech characteristics?

According to the **Source-Filter Model** of speech:

- **Source:** Vocal cords produce raw buzzing pitch (F_0).
- **Filter:** Throat, mouth, tongue, and lips form a resonant filter (vocal tract) producing **formants** (F_1, F_2, F_3).

By taking the logarithm and DCT, MFCCs discard the pitch buzz and capture the **vocal tract shape**. Because every human has a unique physical vocal tract length and shape, MFCC acts as an acoustic fingerprint.

3.5 Advantages and Limitations of MFCCs

Advantages:

1. **Compact:** Compresses thousands of audio samples into just 40–120 numbers per frame.
2. **Noise Tolerant:** Focuses only on speech-relevant frequencies.
3. **Decorrelated:** The DCT step makes the numbers mathematically independent, which speeds up machine learning training.

Limitations:

1. **Loss of High Frequencies:** Mel filters bunch high frequencies ($>4\text{ kHz}$) into broad bins, which can blur subtle AI vocoder artifacts.
2. **No Phase Information:** Discards phase entirely (uses only magnitude $|X[k]|^2$).
3. **Modern AI Clones Match Mel Spectrograms:** Many modern TTS systems are trained specifically to produce realistic Mel spectrograms, making detection harder if using MFCCs alone.

3.6 Comparison: MFCC vs. Spectrogram vs. Mel-Spectrogram vs. Raw Waveform

Representation	Domain	Size / Complexity	Phase Kept?	Human Hearing Scale?
Raw Waveform	Time ($x[t]$)	Huge ($16{,}000$ numbers/sec)	Yes	No
Linear Spectrogram (STFT)	Time + Frequency	High (257 bins/frame)	No (magnitude only)	No
Mel Spectrogram	Time + Mel Frequency	Medium ($64\text{–}128$ bins/frame)	No	Yes
MFCC (Selected)	Quefrency	**Very Compact ($40\text{–}120$ values/frame)**	No	**Yes (Mel + DCT)**

Topic 4: Why Are We Using BiLSTM, FFT, and MFCC Together?

4.1 What is the individual role of each technique?

- **FFT:** Converts raw audio from the time domain to the frequency domain.
- **MFCC:** Compresses that frequency spectrum into a compact representation of the vocal tract shape.
- **BiLSTM:** Takes those frame-by-frame vocal tract shapes and learns how they move forward and backward across time.

4.2 Do FFT and MFCC perform the same task?

No. FFT is an internal calculation step inside the MFCC pipeline.

You cannot calculate an MFCC without first calculating an FFT.

- FFT produces raw frequency bins.
- MFCC applies Mel filters, logarithm, and DCT on top of the FFT bins to produce a clean, compact feature.

4.3 What is the complete data flow from input audio to final prediction?

```
[Raw Audio (16 kHz Mono)]
       ■
       ▼
[Audio Preprocessing (DC removal, 25 dB silence trimming, 0.95 peak scaling)]
       ■
       ▼
[Framing (25 ms Hann window, 10 ms hop) -> 512-point FFT]
       ■
       ▼
[Mel Filterbank (64) + Log + DCT-II + Deltas]
       ■
       ▼
[Input Feature Tensor: (Batch, 301 frames, 120 features)]
       ■
       ▼
[BiLSTM Layer 1 (64 units) + Dropout (0.2)]
       ■
       ▼
[BiLSTM Layer 2 (32 units) + Dropout (0.2)]
       ■
       ▼
[Dense Layer (64 units) + Batch Normalization + L2 Unit Normalization]
       ■
       ▼
[64-Dimensional Unit Voiceprint Vector on Hypersphere]
       ■
       ▼
[Cosine Distance Decision against Enrolled Centroids (Threshold τ = 0.52)]
       ■
   ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
   ▼                              ▼
[Cosine >= 0.52]              [Cosine < 0.52]
Authenticated Speaker       Rejected: Unknown / AI Clone
```

4.4 Is this combination scientifically justified?

Yes. It is a mathematically proven, established pipeline in speech biometrics:

- Raw audio is too high-dimensional to feed directly into small recurrent networks.
- FFT + MFCC reduces dimensionality from $16{,}000$ values/second to a clean matrix of $301 × 120$.
- BiLSTM captures the sequential and temporal dynamics of those features.
- L_2 Hypersphere normalization provides a reliable way to reject unknown/cloned voices.

Topic 5: Alternative Approaches and Their Comparison

5.1 Comparison Table of Approaches

Approach	Feature Used	Temporal Modeling	Strengths	Major Weakness for Our Project
Our Approach (BiLSTM + MFCC)	**120-dim MFCC**	**BiLSTM (Past + Future)**	**Fast, accurate on small datasets, low CPU cost, open-set rejection.**	Loses high-frequency details ($>4\text{ kHz}$) due to Mel scale.
Unidirectional LSTM	MFCC / Mel Spec	LSTM (Past only)	Lower latency, simpler.	Misses future co-articulation context; lower accuracy.

Approach	Feature Used	Temporal Modeling	Strengths	Major Weakness for Our Project
GRU	MFCC / Mel Spec	GRU (Past only)	Trains faster than LSTM (fewer gates).	Misses future context; slightly less expressive.
2D CNN (ResNet)	Spectrogram / Mel Spec	None (Local convolutions)	Excellent at local time-frequency patterns.	Struggles with long-range sequential ordering over $>3\text{ seconds}$.
CNN-LSTM (CRNN)	Mel Spectrogram	CNN for features, LSTM for time	Best of both worlds for audio images.	More complex to tune; larger model size.
Transformer (Conformer)	Mel Spec / Filterbanks	Self-Attention	State-of-the-art on massive datasets.	**Severe data hunger**; overfits on small student datasets; heavy compute.
RawNet2 (Raw Waveform)	Raw time samples $x[t]$	Sinc-Convolutions + GRU	Keeps all phase information; no manual features.	Hard to train; sensitive to background noise; slower.
LFCC (Linear Frequency)	Linear Filterbanks + DCT	Any classifier	**Captures high-frequency vocoder artifacts** (ASVspoof standard).	Slightly lower accuracy for identifying genuine human identities.
CQCC (Constant-Q)	Constant-Q Transform	Any classifier	Excellent frequency resolution at low frequencies.	Slower feature extraction; computationally heavy.

Topic 6: Project Justification, Testing & Evaluation Metrics

6.1 Why is this combination suitable for our final-year project?

1. **Runs on Consumer Hardware:** Trains in $< 15\text{ minutes}$ on a regular student laptop without an expensive GPU.
2. **Stable on Small Datasets:** Does not require thousands of hours of data like Transformers.
3. **100% Explainable & Defensible:** Every step (Hann window, FFT, Mel scale, DCT, BiLSTM gates, Cosine similarity) has a clear mathematical foundation that examiners respect.

6.2 How do we prevent overfitting and data leakage?

- **Strict Recording-Level Partitioning:**

Never split audio at the segment level! If you cut a single recording into ten 3-second slices and randomly assign them to train/test, the model simply memorizes the room echo and background noise.

Correct approach: Full recordings are split into **60% Train, 20% Validation, and 20% Test**.

- **Regularization:** L_2 weight regularization (10^{-4}) on LSTM weights + 20% Dropout.
- **Early Stopping:** Training stops automatically if validation loss does not improve for 12 epochs.

6.3 Which evaluation metrics should we use?

1. **Accuracy:** Segment-level accuracy (87.85%) and recording-level accuracy (100.00%).
2. **Precision, Recall, F1-Score:** Measures false acceptances and false rejections for each speaker individually.
3. **Equal Error Rate (EER):** The point where False Acceptance Rate (FAR) equals False Rejection Rate (FRR).
4. **Cosine Separation Margin (Δ):** The gap between the lowest genuine voice similarity (0.84) and the highest impostor similarity (0.42).

Topic 7: Complete Project Workflow (Step-by-Step)

```
[Step 1: Input Audio]
Capture audio via browser microphone (WebRTC MediaRecorder) or upload a WAV file.

[Step 2: Audio Preprocessing]
- Standardize audio to 16 kHz Mono float32.
- Remove DC electrical offset: y = y - mean(y).
- Trim silent pauses: librosa.effects.trim(top_db=25).
- Peak normalization: Scale maximum amplitude to 0.95 to fix quiet recordings.

[Step 3: Framing]
- Segment audio into 3.0-second analysis windows with a 1.0-second hop.
- Divide each 3-second window into 25 ms sub-frames with a 10 ms hop.

[Step 4: FFT Analysis]
- Run a 512-point FFT on each 25 ms frame to compute the power spectrum.

[Step 5: MFCC Feature Extraction]
- Apply 64 triangular Mel filterbanks.
- Take logarithm of filter energies.
- Apply DCT-II to extract 40 static coefficients.
- Calculate 40 Delta (velocity) and 40 Delta-Delta (acceleration) coefficients.
- Final output matrix per 3-second segment: (301 frames × 120 features).

[Step 6: BiLSTM Processing]
- BiLSTM Layer 1: 64 forward + 64 backward units with 0.2 dropout.
- BiLSTM Layer 2: 32 forward + 32 backward units with 0.2 dropout.
- Dense layer (64 units) + Batch Normalization.
- L2 Unit Normalization projects the vector onto a 64-dimensional unit hypersphere.

[Step 7: Open-Set Rejection]
- Calculate cosine dot products against precomputed enrolled centroids (centroids.npy).
- Decision rule: If highest similarity >= 0.52 and margin >= 0.05, mark as Authenticated.
- If similarity < 0.52, reject as "Unknown Speaker" (Impostor or AI Voice Clone).

[Step 8: Energy-Weighted Voting]
- Aggregate multiple 3-second segment predictions using RMS energy weighting:
  w_i = RMS_i * (1 - 0.5 * P_background).
- Guarantees silent pauses do not corrupt the final identification.

[Step 9: Parallel Forensics & Speech-to-Text]
- Offline Whisper (faster-whisper int8): Transcribes spoken words locally without internet.
- Deepfake Detector (Wav2Vec2 ONNX): Checks pitch dynamics and spectral naturalness.

[Step 10: Presentation Console]
- Web dashboard displays: Identified Speaker, Confidence %, Unknown Warning,
  Interactive MFCC Heatmap, Transcribed Speech, and Downloadable PDF Audit Report.
```

Topic 8: Presentation & Viva Defense Questions (Direct Answers)

Question 1: "Why did you choose BiLSTM instead of a CNN or Transformer?"

Answer: *"Speech articulation is sequential and continuous. When humans speak, the vocal tract prepares for the next sound while articulating the current sound (co-articulation). A Bidirectional LSTM reads both forward and backward, capturing past and future phoneme context. CNNs focus primarily on local image patches and struggle with long-term temporal order. Transformers require massive datasets to train effectively and easily overfit on small student datasets. BiLSTM offers the ideal balance of sequential accuracy, low compute requirements, and high training stability on our hardware."*

Question 2: "Why do you use MFCC for audio feature extraction?"

Answer: *"MFCCs are based on the Source-Filter model of human speech. By taking the DCT of the log Mel spectrum, MFCC separates the vocal tract resonant filter (formants that define a person's biological vocal identity) from the vocal cord pitch buzz. It reduces raw audio from 16,000 samples per second into a compact 120-dimensional feature matrix per frame, preventing our deep learning model from overfitting while retaining the speaker's vocal characteristics."*

Question 3: "Why do you need FFT if you are already using MFCC?"

Answer: *"FFT is not an alternative to MFCC; it is a required internal step within the MFCC extraction process. Audio is recorded in the time domain as pressure waves over time. To apply Mel-scale frequency filters, the audio must first be converted into the frequency domain. We use a 512-point FFT on each 25 ms frame to generate the power spectrum, upon which the Mel filterbanks and DCT are applied. Without FFT, MFCC cannot be computed."*

Question 4: "What is your novel contribution in this project?"

Answer: *"Standard speaker recognition systems use a closed-set softmax classifier, which forces every voice to match an enrolled person—meaning an impostor or cloned voice is misclassified with false high confidence. Our contribution is: 1. An L_2 unit-normalized BiLSTM embedder that maps voices onto a 64-dimensional unit hypersphere. 2. An open-set decision engine using centroid cosine similarity with a calibrated threshold ($τ = 0.52$) and margin ($Δ = 0.05$) to reject unknown impostors and AI clones. 3. An energy-weighted pooling method that achieved 100% recording-level accuracy on our test set. 4. Full integration with 100% offline Whisper transcription and Wav2Vec2 deepfake forensics in a functional web dashboard."*

Question 5: "How does your model distinguish a real human voice from an AI-generated voice?"

Answer: *"Through a dual-layer approach: 1.* ***Biometric Centroid Distance (BiLSTM):*** *If an AI voice is unenrolled or generic, its embedding falls below the cosine similarity threshold ($τ = 0.52$) against enrolled centroids and is rejected as an 'Unknown Speaker'. 2.* ***Acoustic & Neural Forensics (***`detect_deepfake.py`***):*** *In parallel, our Wav2Vec2 classifier and pitch tracking examine micro-discontinuities across frame boundaries, unnatural pitch stability (lack of human jitter), and vocoder high-frequency artifacts."*

Question 6: "What happens if an AI-generated voice sounds almost identical to a human voice?"

Answer: *"Even when a neural clone sounds convincing to human ears, it often leaves mathematical traces in domains human hearing does not perceive well. Human ears are largely phase-deaf and compress frequencies above 4 kHz. Neural vocoders optimize perceptual loss, which frequently leaves artifacts in higher linear frequencies or produces unnaturally rigid pitch trajectories. Our deepfake forensic module inspects these exact mathematical indicators (pitch dynamics and spectral variance) to detect synthetic voices."*

Question 7: "Can your system detect every voice cloning technique?"

Answer: *"No, and scientifically no system can claim 100% detection across all unseen cloning tools. Detection accuracy depends on the vocoder architecture used to create the clone. While our system reliably detects traditional concatenative, parametric, and neural vocoders (like Tacotron, FastSpeech, and HiFi-GAN), newer zero-shot diffusion models present an ongoing challenge. Generalization against completely unseen generative models remains an open research topic in the speech forensics community."*

Question 8: "What are the limitations of your proposed system?"

Answer: *"We identify three clear limitations: 1.* ***Mel Scale Filtering:*** *Because the Mel scale compresses higher frequencies, subtle vocoder artifacts above 6 kHz can be softened. Adding Linear Frequency Cepstral Coefficients (LFCC) in the future will improve this. 2.* ***Extreme Acoustic Noise:*** *Very noisy or reverberant environments can distort speech temporal features. 3.* ***Non-Streaming Latency:*** *Because BiLSTM needs future frames for its backward pass, it requires a 1 to 3 second audio buffer before evaluating."*

Question 9: "How did you verify that your model actually performs well?"

Answer: *"We evaluated our system on a strictly held-out test set partitioned at the recording level (zero data leakage). The results were: -* ***87.85% Segment-Level Accuracy*** *across 288 test segments. -* ***100.00% Recording-Level Accuracy*** *across 44 full test recordings using energy-weighted voting. -* ***0.88 Weighted F1-Score****. -* ***100% rejection rate*** *against unenrolled test voices using our calibrated threshold ($τ = 0.52$) with a separation margin of 0.42 between genuine and impostor scores."*

Question 10: "What improvements will you implement in the future?"

Answer: *"For future work, we plan to: 1. Incorporate **LFCC (Linear Frequency Cepstral Coefficients)** alongside MFCC to directly capture high-frequency vocoder cues based on the ASVspoof standard. 2. Add **in-browser self-enrollment**, allowing new users to record 3 voice samples and dynamically calculate a new centroid without retraining the model. 3. Implement **anti-spoofing replay detection** to detect when a voice is being played back through a phone or loudspeaker. 4. Export the model to **ONNX / TensorRT** for deployment on low-cost edge devices like a Raspberry Pi 5."*

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